

Problem Setup

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Problem Setup

We are given the quadratic equation $x^2 + 2x + 1 = 0$. The objective is to find all values of x that satisfy this equation.

Strategy and Key Principles

To solve a quadratic equation of the form $ax^2 + bx + c = 0$, we can employ several methods, including factoring, completing the square, or using the quadratic formula.

In this specific case, the left-hand side of the equation can be recognized as a perfect square trinomial because it fits the algebraic identity:

$$(x + a)^2 = x^2 + 2ax + a^2$$

Comparing $x^2 + 2x + 1$ to this identity, we find that $a = 1$, which means the expression factors directly into $(x + 1)^2$. We will solve for x by taking the square root of both sides.

Detailed Solution

We begin with the given quadratic equation:

$$x^2 + 2x + 1 = 0$$

Next, we factor the left-hand side of the equation as a perfect square trinomial:

$$(x + 1)^2 = 0$$

To eliminate the exponent, we take the square root of both sides of the equation:

$$\begin{aligned}\sqrt{(x + 1)^2} &= \sqrt{0} \\ x + 1 &= 0\end{aligned}$$

Now, we solve for x by subtracting 1 from both sides:

$$x = -1$$

Final Answer

The solution to the equation is

$$\boxed{x = -1}$$

Key Insights

- **Discriminant Analysis:** The discriminant $\Delta = b^2 - 4ac$ for this equation is $2^2 - 4(1)(1) = 0$. A discriminant of zero confirms that the equation has exactly one distinct real root (or a repeated real root of multiplicity 2).
- **Multiplicity:** Although factoring yields $(x + 1)(x + 1) = 0$, giving $x = -1$ twice, it is conventionally stated as a single unique solution with a multiplicity of two.
- **Efficiency:** Recognizing perfect square trinomials saves considerable time compared to using the full quadratic formula, though the quadratic formula will always yield the same correct result.